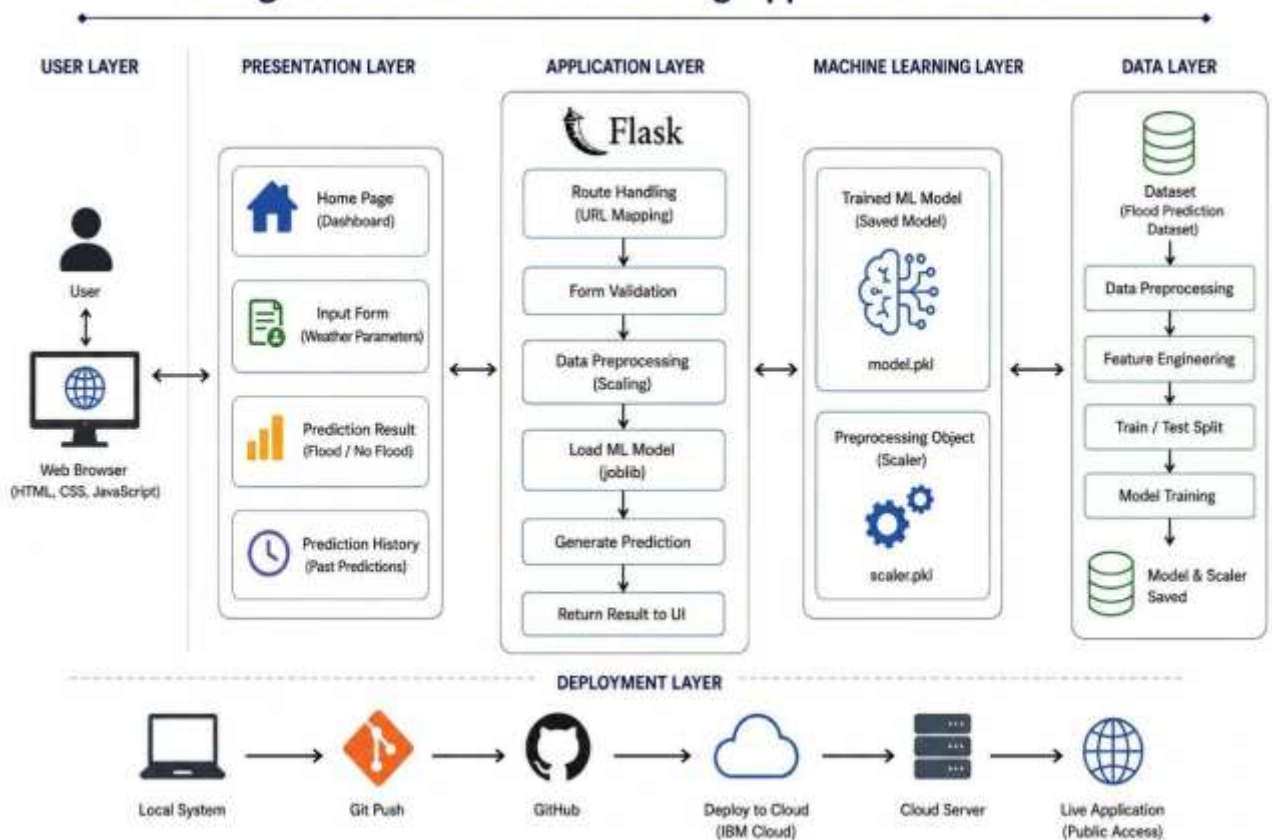


Solution Architecture

Date	28 June 2026
Team ID	
Project Name	Rising water
Maximum Marks	5 Marks

Solution Architecture Diagram:

Rising Waters: A Machine Learning Approach to Flood Prediction



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation / Client Layer	The client-side interface where users access the flood prediction dashboard, input weather parameters, and view their historical predictions.	HTML5, CSS3, JavaScript, Jinja2
API Gateway / Load Balancer	Serves as the entry point for internet traffic, securely routing incoming HTTP requests directly to the Flask web application.	Render (Cloud Managed Proxy)
Auth Service	Secures application routes by authenticating users upon login, mathematically hashing passwords, and maintaining secure sessions via encrypted cookies.	Flask Sessions, Werkzeug Security
Core Logic Service	The central backend logic that processes API requests, validates inputs, executes data preprocessing, and runs the machine learning model inference.	Python, Flask, XGBoost, Scikit-learn, Joblib
ExternalAPI Integrations	The application communicates securely over the network to external cloud-managed services rather than relying on local environments.	Neon.tech (Cloud PostgreSQLAPI)
Data / Storage Layer	Provides highly reliable, scalable relational data storage to persistently save user credentials and all past flood prediction records.	PostgreSQL, SQLAlchemy (ORM)